SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Global Dark and Light Mode Theme Implementation

Field	Details
Change Request ID	CR-012
Change Request Title	Global Dark and Light Mode Theme Implementation
Student Name	T.Manoja
Student ID	JF_ICT_24_12
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System previously used fixed interface colors. A dark/light theme option is intended to improve readability and user comfort while keeping the appearance consistent across the main application pages.

The change request covers the Home/Landing page, Login page, Administrator Dashboard, and Student Dashboard. The selected theme should remain consistent while navigating between these pages and, where implemented, after a browser refresh.

1.2 Existing Behaviour

- The interface used page-specific or fixed colors.
- A single theme could not be selected consistently across all main pages.
- Some backgrounds, text, forms, and table elements needed theme-aware styling.
- The selected theme needed to remain consistent during navigation.

1.3 Requested Change

- Provide a control to switch between light mode and dark mode.
- Apply theme-aware colors to Home, Login, Admin Dashboard, and Student Dashboard.
- Update backgrounds, text, cards, forms, tables, borders, and related interface elements.
- Keep the theme consistent when navigating between pages.
- Persist the theme preference with browser local storage if included in the implementation.
- Preserve authentication, authorization, student data, APIs, and existing CRUD operations.

2. Implementation

2.1 Implementation Summary

The theme feature is a frontend presentation enhancement. The implementation plan uses shared theme state and theme-aware CSS so the selected appearance can be applied consistently across the application. Theme switching should not alter login credentials, user roles, student records, or backend business logic.

Screenshots in Section 5 provide visual evidence of the Home page, Login page, Administrator Dashboard, and Student Dashboard in the available theme states.

2.2 Affected Files

File / Location	Change	Purpose
frontend/src/App.jsx or frontend/src/main.jsx	Verify actual project entry point	Ensure the shared theme provider wraps the application routes, if used.
frontend/src/context/ThemeContext.jsx	Add / modify if used	Manage the current theme and theme-switching function.
frontend/src/components/ThemeToggle.jsx	Add / modify if used	Provide the reusable light/dark toggle control.
frontend/src/components/Navbar.jsx	Modify if applicable	Display the toggle in the relevant navigation areas.
frontend/src/styles/index.css or global.css	Modify	Define shared theme variables and global styles.
frontend/src/styles/landing.css	Modify / verify	Apply theme-aware styling to the Home/Landing page.
frontend/src/styles/login.css	Modify / verify	Apply theme-aware styling to the Login page.
frontend/src/styles/dashboard.css	Modify / verify	Apply theme-aware styling to the Admin Dashboard.
frontend/src/styles/studentDashboard.css	Modify / verify	Apply theme-aware styling to the Student Dashboard.

2.3 Technical Changes

2.3.1 Theme Management

A shared theme state can be provided through React Context so pages use the same selected theme rather than maintaining unrelated page-level values.

2.3.2 Theme Toggle

The toggle changes the selected theme between light and dark mode and should be available on the required pages.

2.3.3 Theme-Aware CSS

CSS custom properties or equivalent theme-specific rules can control primary and secondary backgrounds, text, borders, input fields, cards, table headers, hover states, and shadows.

2.3.4 Theme Persistence
If local-storage persistence is enabled in the code, the application can save the selected theme and restore it on startup. Confirm this behaviour with a refresh test before marking it verified.

2.3.5 Cross-Page Consistency
The same selected theme should be reflected on Home, Login, Administrator Dashboard, and Student Dashboard. Any page elements that remain fixed to a single color should be reviewed for contrast and consistency.

2.4 Implementation Logic
1. Initialize the selected theme when the application loads.
2. Make the theme state available to relevant pages and components.
3. Allow the user to switch between light and dark modes.
4. Apply the corresponding theme styles to the page interface.
5. Save and restore the preference if persistence is part of the implementation.
6. Check theme consistency during navigation and refresh.
7. Regression-test login and student CRUD operations to ensure they are unaffected.

2.5 Error and Data Handling
The theme feature should remain a frontend-only change. It should not require database schema changes or new backend endpoints. Authentication, authorization, student records, and CRUD operations should continue to use their existing logic. This report does not claim those regression tests are proven by screenshots alone.

3. Testing & Verification

3.1 Testing Approach
Use manual UI and regression testing in the development environment. Record the actual result after performing each test; do not mark a test as passed unless it has been executed.

Environment: Development
Browser: [Enter browser and version]
Test Roles: Guest, Administrator, Student

3.2 Test Cases

Test ID	Test Case	Expected Result
TC-01	Open Home page	Page renders correctly in the default theme.
TC-02	Switch light to dark	Theme changes and text remains readable.
TC-03	Switch dark to light	Light theme is applied correctly.
TC-04	Open Login page	Login page follows the selected theme.

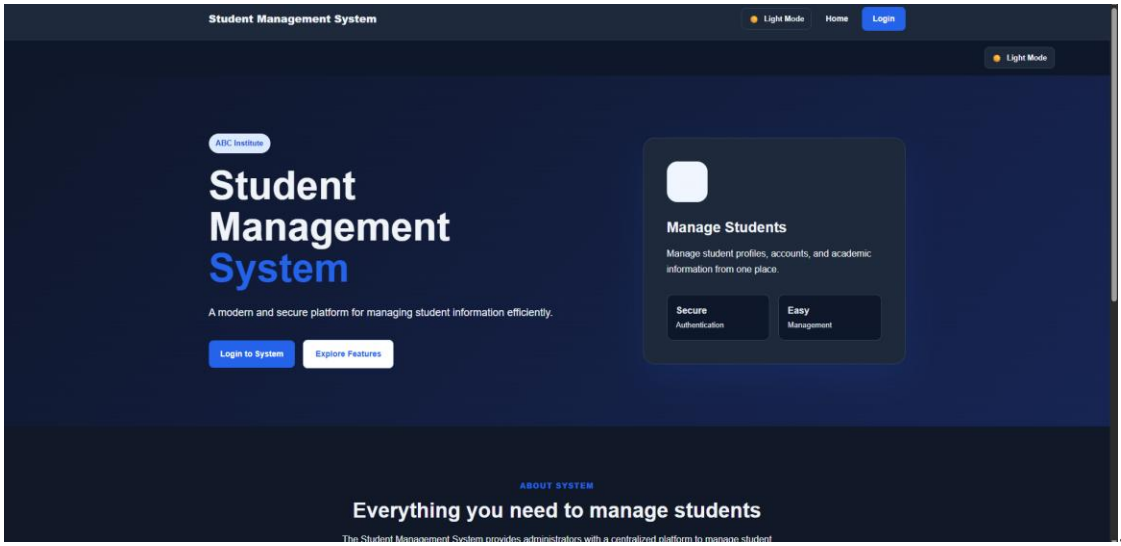
TC-05	Open Admin Dashboard	Dashboard, forms, table, and text follow the theme.
TC-06	Open Student Dashboard	Student dashboard follows the theme.
TC-07	Navigate between pages	Selected theme remains consistent.
TC-08	Refresh the browser	Saved theme is restored, if persistence is implemented.
TC-09	Check text and input contrast	Text, labels, and inputs remain readable.
TC-10	Login and logout	Existing authentication behaviour is unchanged.
TC-11	Student CRUD operations	Create, read, update, and delete still work.
TC-12	Check API/database behaviour	Theme switching does not modify student data or APIs.

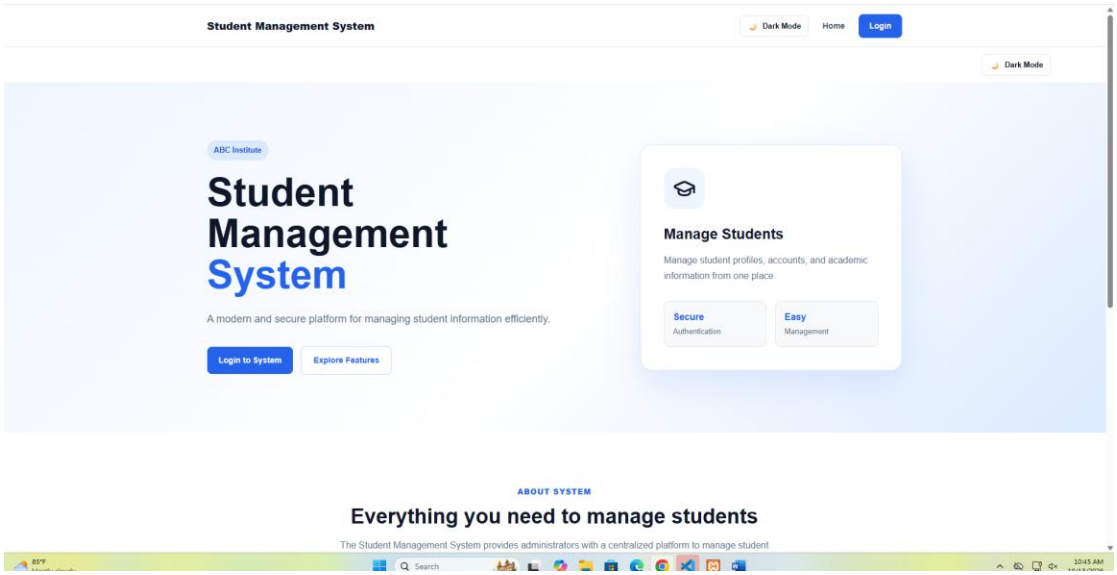
4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Users can switch between light and dark themes.	Toggle test	Visual evidence / verify both directions
AC-002	Theme applies across major pages.	Inspect Home, Login, Admin, Student pages	Screenshots available
AC-003	Text and interface elements remain readable.	UI contrast inspection	Review all visible elements
AC-004	Theme persists after browser refresh.	Refresh test	Not established by screenshots alone
AC-005	Theme stays consistent during navigation.	Navigate between pages	Requires live verification
AC-006	Authentication and student CRUD remain unchanged.	Regression tests	Requires live verification
AC-007	No backend or database change is needed for theme switching.	Code review	Expected frontend-only scope; verify implementation

5. Evidence

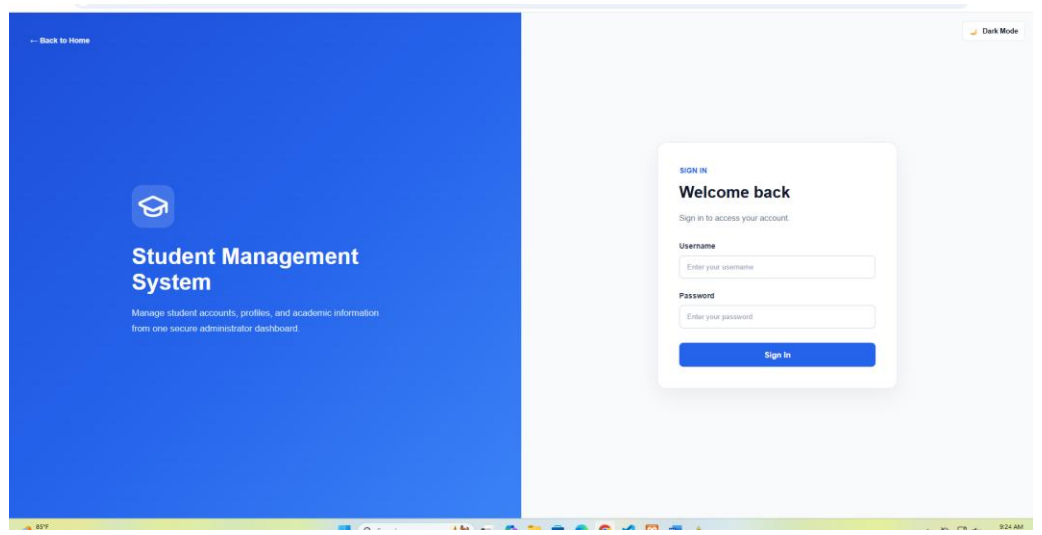
Evidence 01 – Home / Landing Page

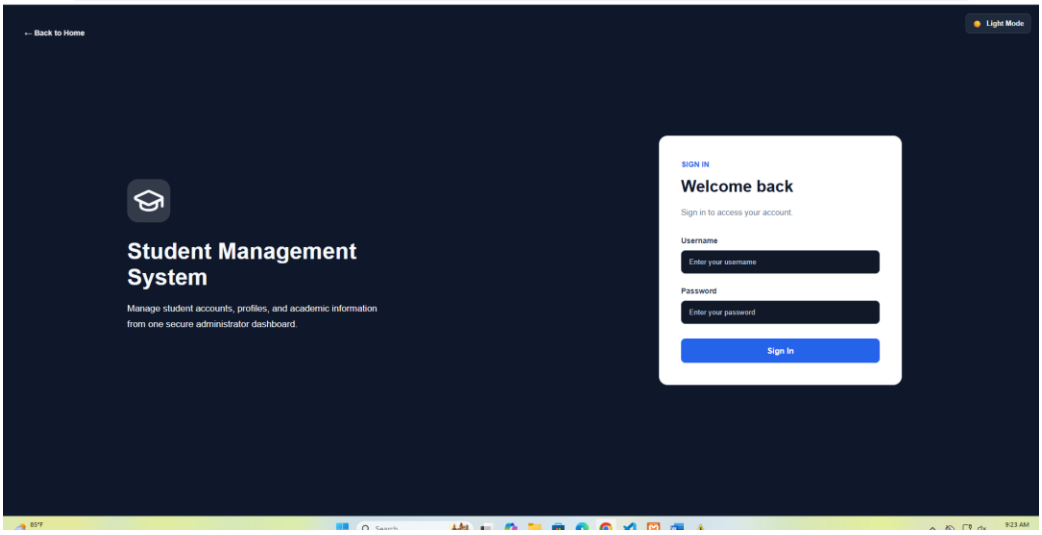




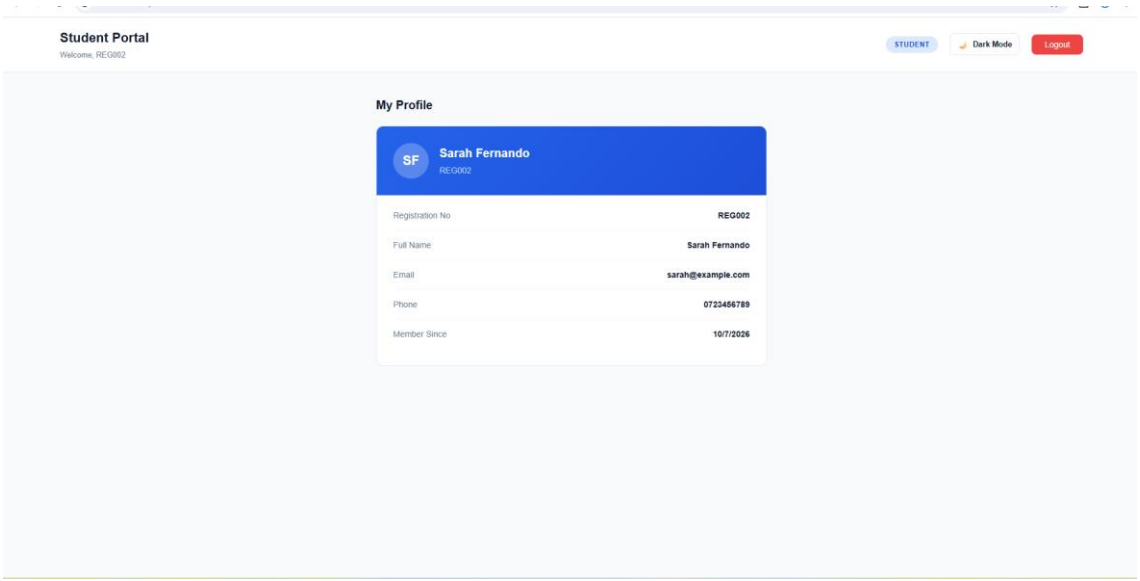
Evidence 02 – Login Page

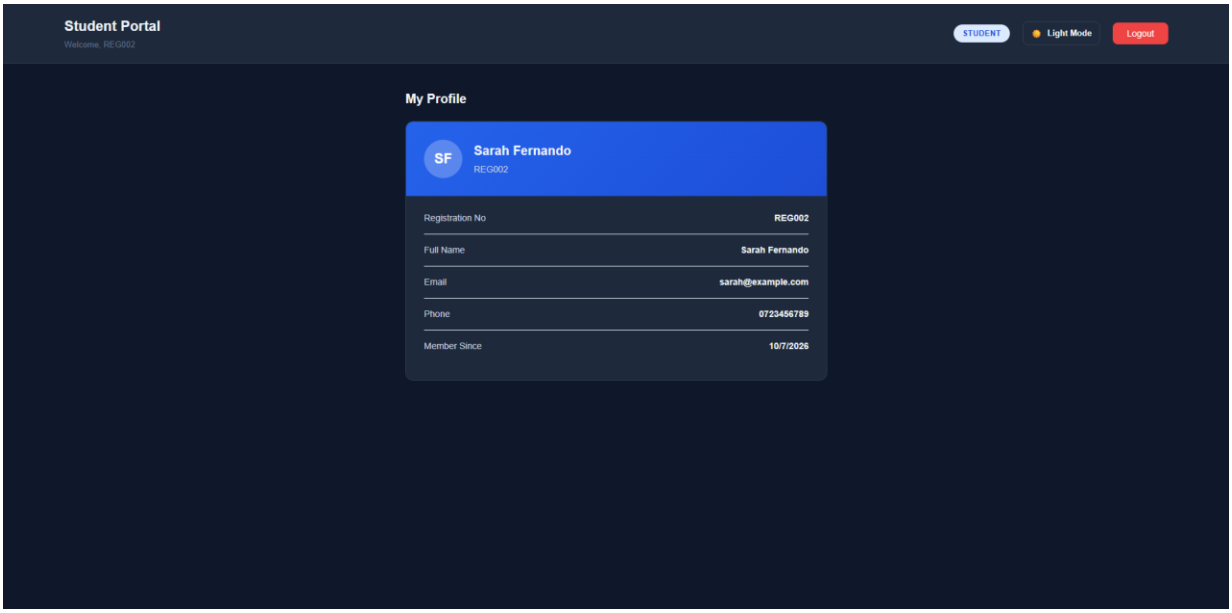
Description: Screenshot evidence of the Login page and theme control.



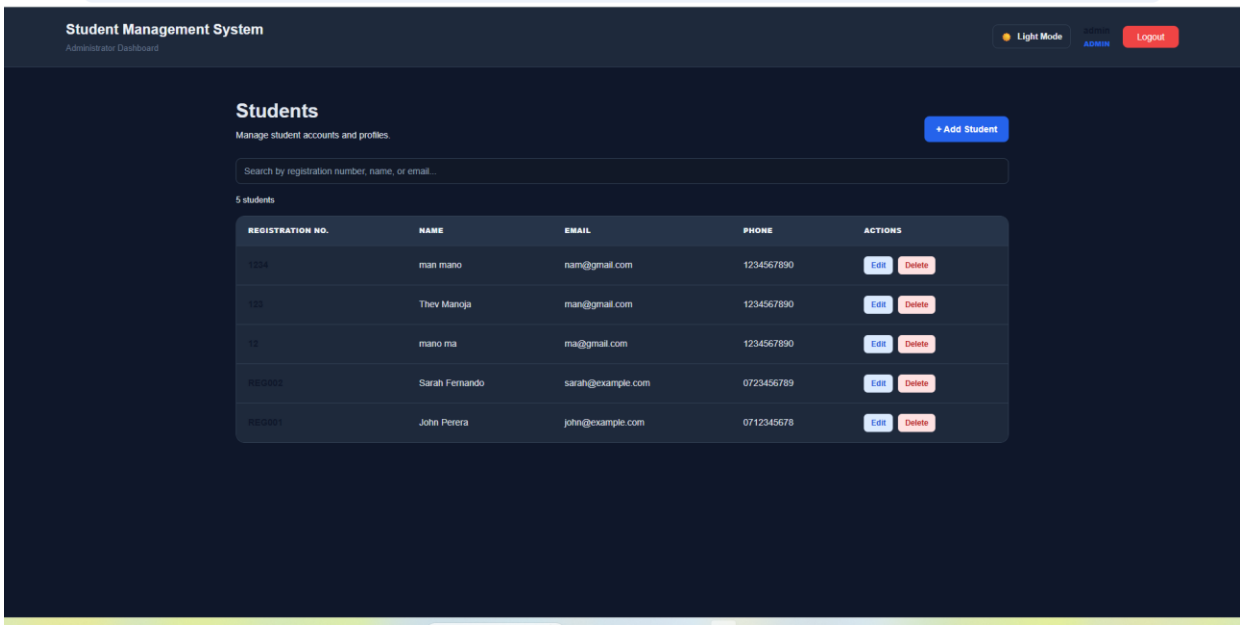


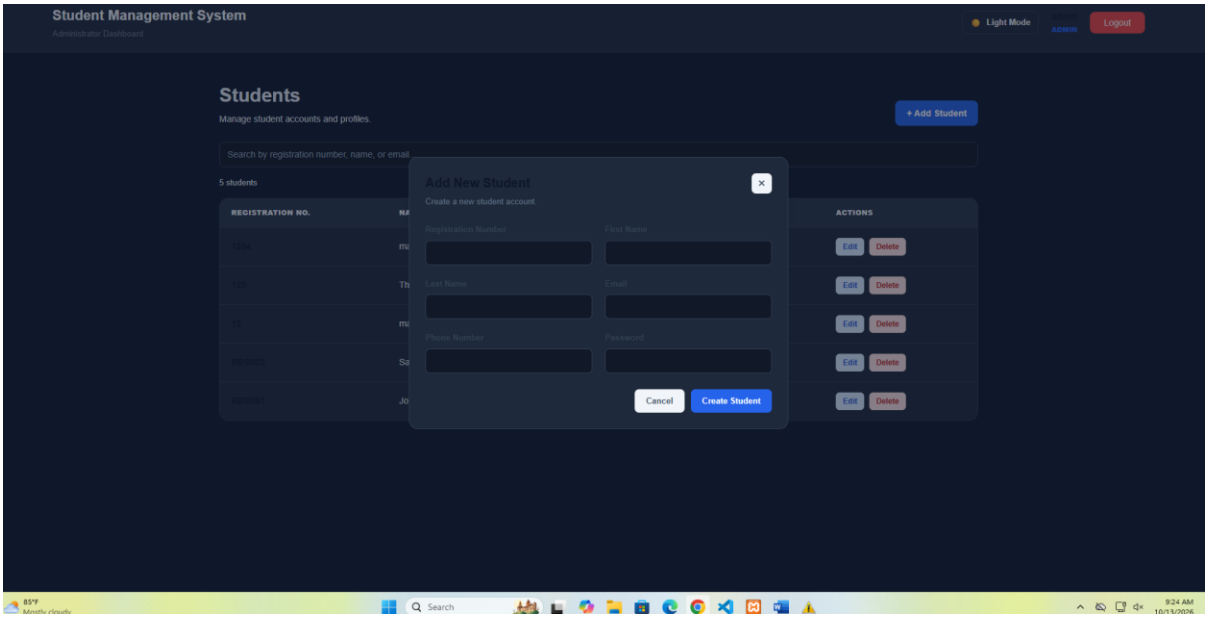
Evidence 03 – Student Dashboard
Description: Screenshot evidence of the Student Dashboard in dark mode.





Evidence 04 – Administrator Dashboard
Description: Screenshot evidence of the Administrator Dashboard in dark mode.





6. Final Result

Item	Result
Change scope	Global frontend theme enhancement
Evidence included	Home, Login, Student Dashboard, Admin Dashboard
Backend/API/database change	Not intended for this theme-only request
Authentication and CRUD	Must remain unchanged; regression test required
Final status	Implementation report prepared; verify outstanding tests before claiming full acceptance

6.1 Final Summary

CR-012 introduces a dark/light theme capability for the Student Management System. The report documents the intended theme-management approach, affected frontend areas, test cases, acceptance

criteria, and screenshot evidence from the available Home, Login, Administrator Dashboard, and Student Dashboard captures. Before final submission, confirm the exact changed files in the project and execute the remaining persistence, navigation, authentication, and CRUD regression tests.

6.2 Issues / Limitations

The screenshots show the interface at specific points in time and cannot alone establish that the theme persists after refresh or that unrelated backend functionality remains operational. Replace the name and student ID placeholders and update test statuses based on tests actually performed.

Student Name	Student ID	Date
T.Manoja	JF/ICT/24/12	13 October 2026